

Nina M. Brown

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Education

Ph.D. in Astronomy and Astrophysics, University of Arizona | Tucson, AZ | 2031 (expected)

Graduate Student-at-Large, University of Chicago | Chicago, IL | 2023

B.S. in Astrophysics, University of Florida | Gainesville, FL | May 2022 | Summa Cum Laude

Research Experience

Graduate Research Assistant, PI Dr. Chad Bender | Aug. 2025 - present | U. of Arizona | 20 hr/wk

- Characterizing the performance of CSHELL (nIR/IR spectrograph).
- Upgrading CSHELL components in a cleanroom environment.
- Updating the CSHELL data reduction pipeline from IDL to Python.

MAROON-X Research Tech, PI Dr. Jacob Bean | Dec. 2023 - Aug. 2025 | U. of Chicago | 40 hr/wk

- Reduced, analyzed, and distributed MAROON-X (optical/nIR RV spectrograph) data products.
- Assisted program PIs in reading and interpreting data products.
- Took calibration and science spectra with MAROON-X, installed at the Gemini North Telescope.
- Led a project to characterize the planetary system of TOI-2142.

Research Specialist II, PI Dr. Laurens Mets | Aug. 2022 - Nov. 2023 | U. of Chicago | 40 hr/wk

- Studied microbial gas-exchange reaction efficacy via Python simulations.
- Programmed and operated computer interfacing for lab instruments using serial controls.
- Prepared and maintained a stirred-tank reactor in a wet lab.

Undergrad Research Assistant, PI Dr. Paul Fulda | May 2021 - May 2022 | U. of Florida | 10 hr/wk

- Studied higher-order Hermite-Gauss beam mode efficacy via Python simulations.
- Tested simulation results on an optics bench.

Undergrad Research Assistant, PI Dr. Zach Slepian | Mar. 2020 - Jan. 2022 | U. of Florida | 10 hr/wk

- Studied the relative velocity between baryonic and dark matter in the early Universe, using large data sets from the 2MASS Redshift Survey, analyzed through Python.

Outreach and Mentorship

TIMESTEP Graduate Mentor, Apr. - Sept. 2026 | U. of Arizona | Tucson, AZ | 3 hr/wk

- Supporting and guiding undergraduate STEM students through summer internships.
- Facilitating paperwork and interactions between TIMESTEP, students, and companies.

Telescope Facilitator, volunteer | Aug. 2023 - Aug. 2025 | Adler Planetarium | Chicago, IL | 2 hr/wk

- Operated the Doane Observatory 24" telescope, as well as 6-8" telescopes, for public observing.
- Taught guests about the selected target, general observational astronomy, and the Adler planetarium; during 2-3 minute interactions; averaging 200 guests per shift.

Technical Skills

Observational | optical, nIR, IR spectrograph instrumentation; casual to professional observing.

Computational | Python (JupyterLab, VS Code), IDL, Git/Github; LaTeX; certified in SOLIDWORKS.

Operating Systems | Windows, macOS, Linux (Ubuntu).

Selected Publications and Presentations

Brown N., Bender, C., Monson, A., Maekawa, K., O. T. Choi, S., Berkson, J., Schwab, C., Prato, L., Angel, R., *The Large Fiber Array Spectroscopic Telescope: CSHELL revitalization and characterization*. Poster at SPIE Astronomical Telescopes + Instrumentation; July 2026; Copenhagen, Denmark.

Brady, M., Bean, J. L., Basant, R., **Brown, N.**, Das, T., Nixon, M., Luque, R., Piaulet-Ghorayeb, C., Radica, M., Seifahrt, A., Stürmer, J., Zhao, L., 2025, "An Earth-like Density for the Temperate Earth-sized Planet GJ 12b" *Accepted in AJ*, *arXiv: 2506.20561*.

Basant, R., Luque, R., Bean, J. L., Seifahrt, A., Brady, M., Zhao, L., **Brown, N.**, Das, T., Stürmer, J., Kasper, D., Gupta, R., Stefánsson, G., 2025, "Four sub-Earth planets orbiting Barnard's Star from MAROON-X and ESPRESSO" *Accepted in ApJL*, *arXiv: 2503.08095*.

Tao, L., **Brown, N.**, Fulda, P., 2023, "Beam displacement tolerances on a segmented mirror for higher- order Hermite-Gauss modes" *Journal of Optics*, 26:015603, *arXiv: 2305.03681*.

Brown N., Song, A., Slepian, Z., Hou, J., Craigie, M., Eisenstein, D., *Relative Velocity of Dark and Baryonic Matter Within the Milky Way*. Poster at the U. of Florida Spring 2022 Undergraduate Research Symposium; April 2022; Gainesville, FL.